PWM AO4410 2-Channel Output Module

DEVLAB FACTOR

Mfr. Part #: UE0083

Product Reference Manual (V1.0)

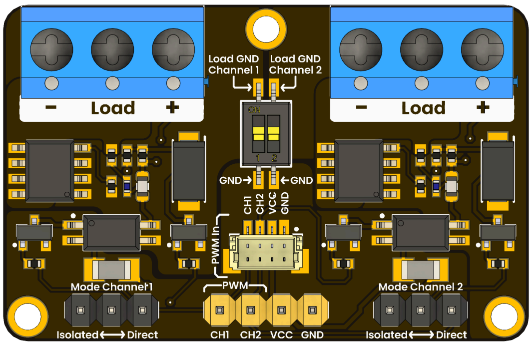
Description

The **DevLab PWM AO4410 2-Channel Output Module** is a compact driver board designed to amplify pulse-width modulation (PWM) signals from a microcontroller, enabling control of higher voltage and current loads with precision and reliability. Each channel is powered by an **AO4410 MOSFET**, providing efficient switching with low heat generation and minimal voltage drop. This makes the module ideal for driving motors, LEDs, solenoids, and other power-demanding devices that require fine control through PWM.

Featuring **clearly labeled screw terminals** for secure load connections and a **4-pin JST 1.0 mm connector** for signal and power input, the module ensures quick and reliable integration with controller boards or external systems. Although the connector is mechanically compatible with Qwiic cables, it does **not** use the Qwiic pinout nor I²C communication. Only the GND and VCC positions follow the conventional layout. Compact, efficient, and easy to integrate, this dual-channel module is ideal for motor control, lighting systems, and other applications requiring precise and stable PWM power regulation.

DevLab format compatibility.

Simplicity and compatibility are the core principles of the **DevLab form factor**. This standard defines a **compact and communication-optimized board layout**, ensuring straightforward connection and interoperability among DevLab module



Key Features

- Dual-channel PWM output stage employing AO4410 N-channel MOSFETs
- Supports high-current loads with a low $R_{DS(on)}$ the purpose of efficient transitioning
- Compliant with logic-level PWM signals at 3.3 V and 5 V
- 4-pin JST connector for rapid signal and power input.
- Screw terminals for secure and effortless cargo connections
- Independent channel control for solenoid, LED, or motor applications
- Optimized for embedded and prototyping environments, the compact PCB layout

Hardware Features

- **2 × AO4410 N-channel MOSFETs** rated for high current and low $R_{DS(on)}$ operation
- **2 × Independent PWM input channels** for separate load control
- **1 × 4-pin JST connector** (VCC, GND, PWM1, PWM2) for signal and power input
- **2 × Screw terminal blocks** for external load connections (CH1 and CH2)
- **Onboard power indicator LED** for quick visual status (varies by revision)
- **Operating voltage:** 3.3 V – 12 V typical
- **Logic-level compatible inputs** for direct connection to microcontrollers (e.g., ESP32, STM32, Arduino, etc.)
- **Compact PCB design** with labeled pinout for easy integration in embedded systems
- **Thermally optimized MOSFET layout** for stable switching and reduced heat buildup

Software Support

- **Arduino IDE**
- **Micro Python**

Applications

- Motor speed and direction control for DC motors
- Brightness control for high-power LEDs
- Temperature control for heating elements
- Efficient power regulation for various loads
- Amplifying PWM signals for long-distance transmission

CONTENTS

1 The Board

1.1 Accessories

- 2x Jumper bridge
- 1x Single-row male header (1x4)

2 Ratings

2.1 Recommended Operating Conditions

Symbol	Description	Min	Typ	Max	Unit
Vin	Input voltage to power on the module	3.3	-	5	V
V Load*	Maximum voltage for the external load	-	-	30	V
I Load*	Maximum current for the external load	-	-	15	A
P*	Maximum power at load	-	-	200	W

- A DIP switch is provided to couple or isolate the microcontroller ground from the load ground.
- **Mode selection**:** use jumper bridges to choose "Direct mode" (grounds coupled) or "Isolated mode" (grounds isolated) for improved protection.

***These values were tested at 1 kHz**. Performance may vary at other frequencies.
**Use the jumper bridges and the DIP switch to change between Direct and Isolated modes.

3 Functional Overview

The **PWM AO4410 2-Channel Output Module** operates as a dual high-current PWM amplifier, allowing low-power PWM signals from a microcontroller to safely and efficiently control high-power external loads. Each channel functions independently and is based on a **MOSFET switching stage**, enabling fast switching, high efficiency, and reliable thermal performance.

3.1 Signal Flow

1. The microcontroller generates a PWM signal (from a GPIO pin).
2. The signal is fed into the module through:
    - The **JST 1 mm connector**
    - The **4-pin header**.
3. The PWM signal drives the **AO4410 MOSFET** on the selected channel.
4. The MOSFET switches the external power supplied to the load through the **screw terminal block**.
5. The connected load (motor, LED, heater, etc.) responds proportionally to the PWM duty cycle.

This allows precise control of **speed, brightness, or power level** depending on the application.

3.2 Dual-Channel Independent Operation

- The module provides **two fully independent PWM output channels**.
- Each channel has:
    - Its own PWM control input
    - Its own power input
    - Its own load output
    - Its own activity indicator LED
- Both channels can operate at the same or different duty cycles without interference.

3.3 Ground Isolation and Protection

The module includes a **DIP switch and jumper system** that allows selecting between two operational modes:

- **Direct Mode (Ground Coupled)**
    - The microcontroller ground and load ground are connected.
    - Recommended for:
        - Low-noise environments
        - Short cable runs
        - Shared power systems

- **Isolated Mode (Ground Isolated)**
    - The microcontroller ground is electrically isolated from the load ground.
    - Reduces:
        - Ground loops
        - Electrical noise
        - Risk of damage from transient voltages
    - Recommended for:
        - High-current systems
        - Motors with electrical noise
        - Industrial environments

3.4 Power Handling Behavior

- The module **does not power the load directly**.
- External voltage must be applied to the **V Load terminals**.
- The onboard MOSFETs act only as **high-speed electronic switches**.
- Maximum supported:
    - **30 V load voltage**
    - **15 A per channel**
    - **200 W total power**

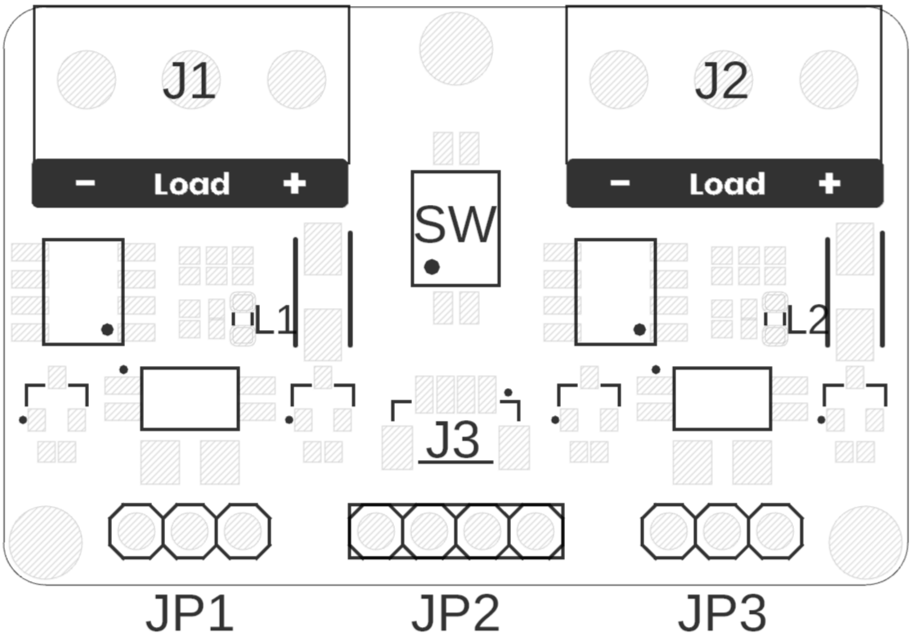
3.5 Visual Status Indication

- Each channel includes a **dedicated indicator LED**:
    - LED ON → PWM signal active
    - LED OFF → No PWM signal
- This allows quick visual verification of:
    - PWM signal presence
    - Channel activation state

3.6 Typical Operating Behavior

- **Low duty cycle (10–30%)** → Low motor speed / dim light
- **Medium duty cycle (40–70%)** → Moderate power output
- **High duty cycle (80–100%)** → Maximum output power

3.2 Board Topology



Top View of Board Topology

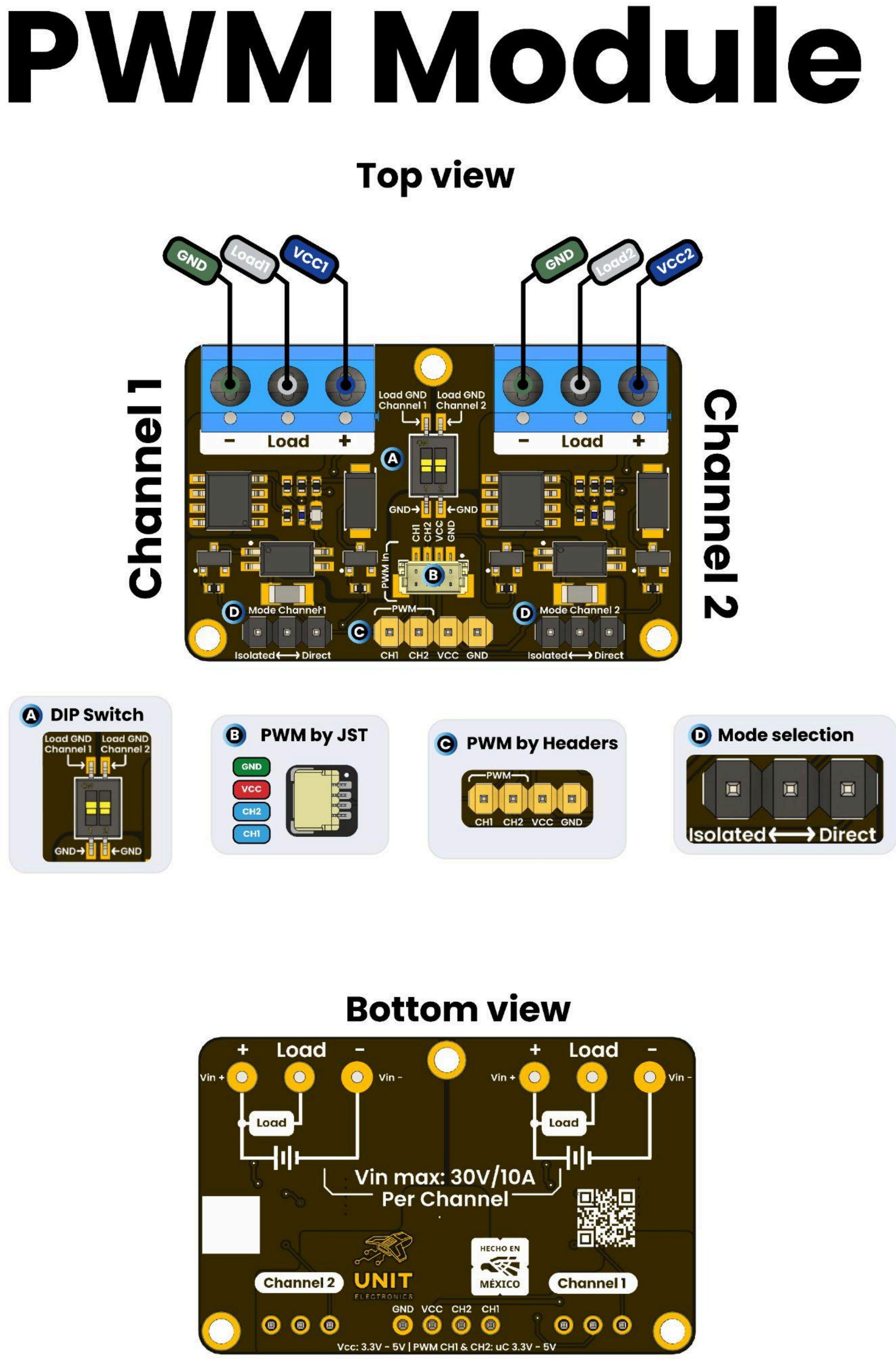
Views of PY32F003L24D6TR DevLab Topology

Table 3.2.1 - Components Overview

Ref.	Description
SW	Dip Switch for coupling grounds
L1	Channel 1 PWM LED
L2	Channel 2 PWM LED
J1	Screw Terminal Block for Channel 1 Load
J2	Screw Terminal Block for Channel 2 Load
J3	JST 1mm Connector for input signals
JP1	Header for Channel 1 mode selection
JP2	Header for input signals
JP3	Header for Channel 2 mode selection

4 Connectors & Pinouts

4.1 General Pinout



PY32F003L24D6TR DevLab *General Pinout*

4.2 Pinout General Description

Channel	Description	Control Pins	Power Pins	Load Terminals	Typical Use
PWM Channel 1	MOSFET-based driver that amplifies MCU's PWM output to switch a heavier external load	PWM1, GND	VCC1, GND	Load1	DC motors, high-power LEDs, solenoids
PWM Channel 2	Identical high-current PWM driver on a second channel	PWM2, GND	VCC2, GND	Load2	Same as Channel 1

5 Board Operation

5.1 Getting Started with arduino IDE

This Arduino IDE example shows basic PWM control by smoothly increasing and decreasing an LED's brightness.

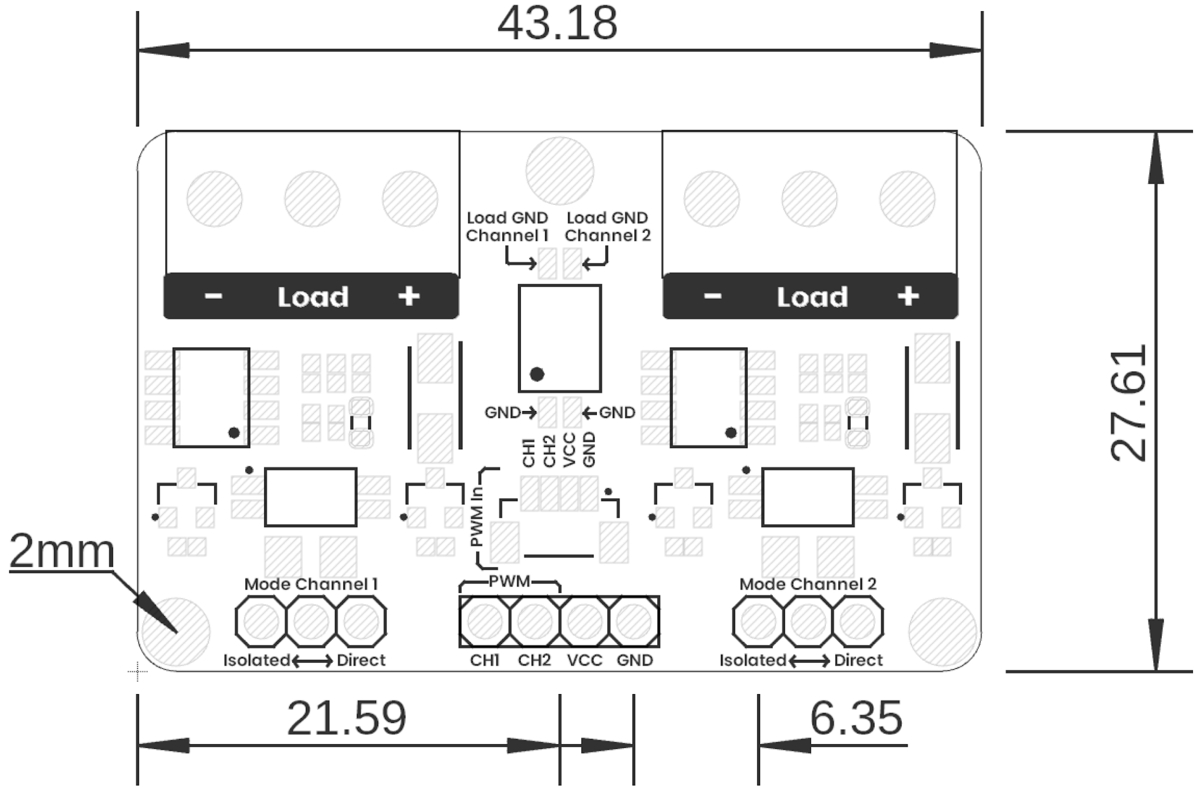
It configures a PWM output pin, varies the duty cycle, and prints the values to the Serial Monitor. The sketch uses `analogWrite()`, allows adjustable brightness steps, and implements a simple fade-in/fade-out pattern.

Hardware: connect the PWM output pin (default: pin 9) to the module's PWM input, attach the LED or load to the output terminals, and ensure proper power connections.

Usage: open the example, select board and port, upload, open Serial Monitor at 115200 baud, and observe the LED fading.

```
// LED Fade + PWM duty print on Serial Monitor

int LED_PIN = 9;       // PWM pin
int STEP    = 5;       // Brightness increment per step

void setup() {
 pinMode(LED_PIN, OUTPUT);
 Serial.begin(115200);          // Match this baud rate in the Serial
Monitor
 delay(200);                    // Small pause to avoid garbage at
start
 Serial.println("Starting fade...");
}

void loop() {
 // Fade up
 for (int duty = 0; duty <= 255; duty += STEP) {
   analogWrite(LED_PIN, duty);
   Serial.print("Duty: ");
   Serial.println(duty);
   delay(500);
 }

 // Fade down
 for (int duty = 255; duty >= 0; duty -= STEP) {
   analogWrite(LED_PIN, duty);
   Serial.print("Duty: ");
   Serial.println(duty);
   delay(500);
 }
}
```

6 Mechanical Information



Board Dimensions in Millimeters

Mechanical dimensions in millimeters

7 Company Information

Company name	UNIT Electronics
Company website	https://uelectronics.com/
Company Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX

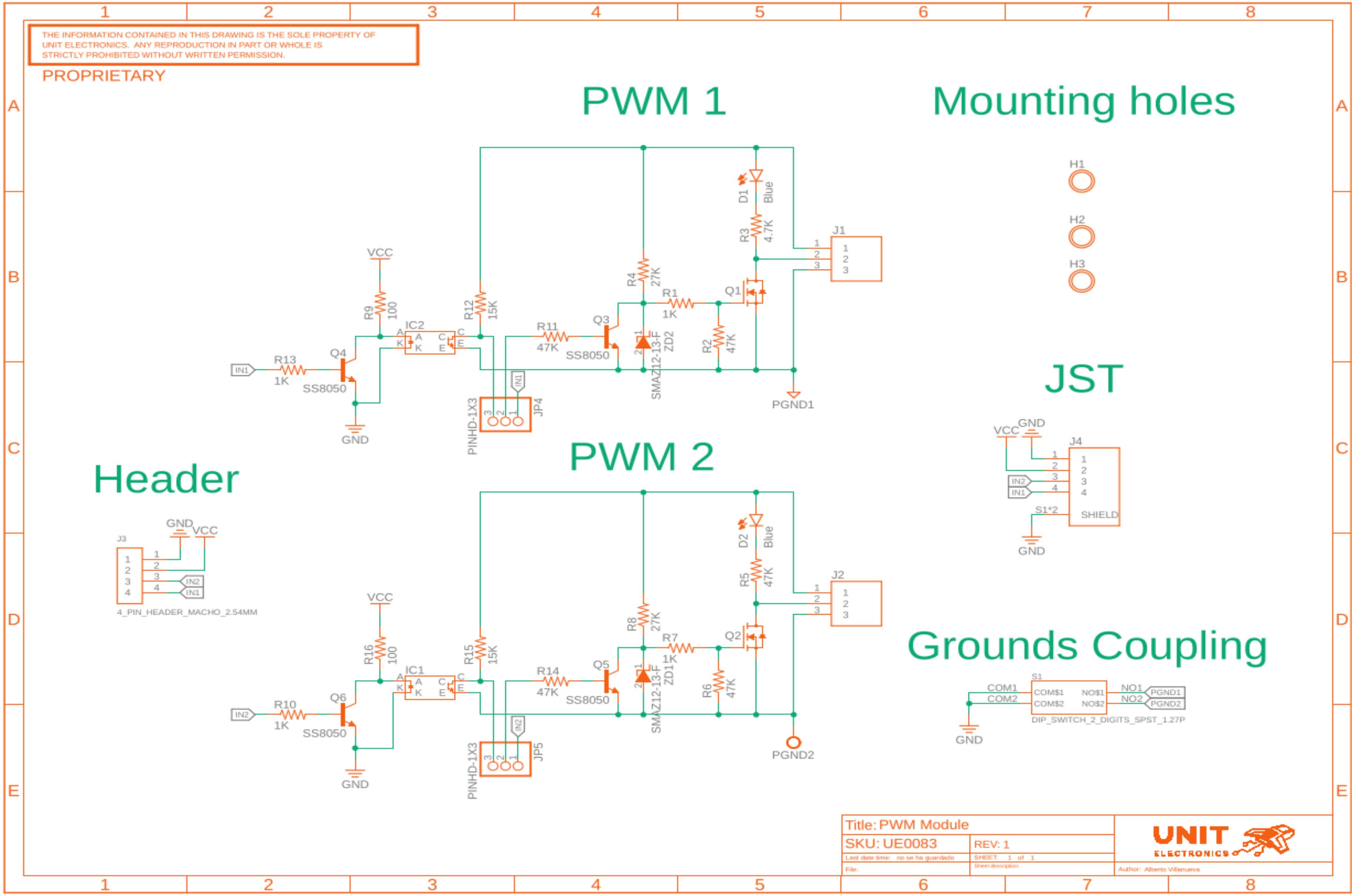
8 Reference Documentation

Ref	**Link**
Wiki	https://wiki.uelectronics.com/wiki/devlab-pwm-ao4410-2-channel-output-module
Github	https://github.com/UNIT-Electronics-MX/unit_devlab_pwm_ao4410_2_channel_output_module

9 Appendix

9.1 Schematic

(https://github.com/UNIT-Electronics-MX/unit_devlab_pwm_ao4410_2_channel_output_module/blob/main/hardware/unit_sch_v_0_0_1_ue0083_pwm_module.pdf)



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PROPRIETARY
PWM 1
Mounting holes
H1
H2
H3
JST
Header
PWM 2
Grounds Coupling
J1
J2
J3
J4
JP4
JP5
Q1
Q2
Q3
Q4
Q5
Q6
SS8050
IC1
IC2
D1
D2
Blue
ZD1
ZD2
SMAZ12-13-F
PINHD-1X3
VCC
GND
PGND1
PGND2
IN1
IN2
4_PIN_HEADER_MACHO_2.54MM
SHIELD
S1
COM1
COM2
NO1
NO2
DIP_SWITCH_2_DIGITS_SPST_1.27P
Title: PWM Module
SKU: UE0083
REV: 1
SHEET: 1 of 1
Author: Alberto Villanueva